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Overview

This problem statement is asking to use supervised machine learning to predict whether a person has heart disease or not. The output, which is whether the person has heart disease or not, is already known and the machine learning algorithm can be trained to make accurate predictions about a person's heart disease status.

Data Preparation

Then, we split the data into train and test sets for model development. Afterward, we implemented the model with the help of sklearn and matplotlib libraries. Finally, we evaluated the model using various metrics like accuracy and precision.

Validation data results

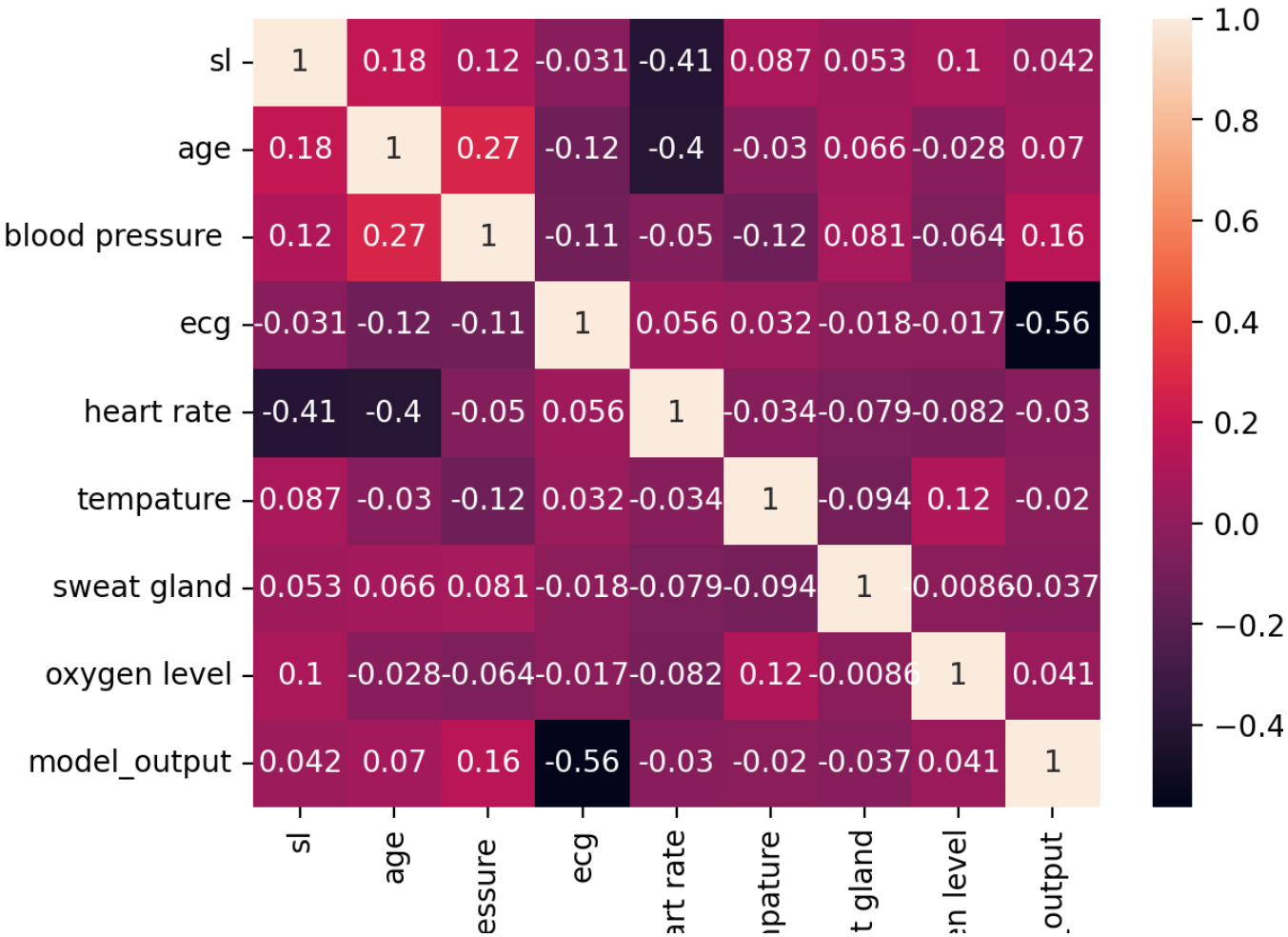
Below is our dataframe that contains the model output along with its confidence score

	sl	age	blood pressure	ecg	heart rate	tempature	sweat gland	oxygen level	model_o
0	0	63	145	0	150	98.2	519	86	
1	1	37	130	1	187	97	520	84	
2	2	41	130	0	172	99.2	532	82	
3	3	56	120	1	178	99	523	86	
4	4	57	120	1	163	98.1	524	89	
5	5	57	140	1	148	97.5	516	87	
6	6	56	140	0	153	97.1	519	82	
7	7	44	120	1	173	98	510	88	
8	8	52	172	1	162	98.6	517	87	
9	9	57	150	1	174	98.7	521	83	

Data stats

	sl	age	blood pressure	ecg	heart rate	tempature	sweat gland	oxygen level	mo
count	299	299	299	299	299	299	299	299	
mean	149	54.3344	131.6656	0.5251	149.7692	99.0853	515.204	90.1639	
std	86.4581	9.0899	17.5958	0.5264	22.8989	1.791	5.7219	5.2003	
min	0	29	94	0	71	96.4	502	78	
25%	74.5	47.5	120	0	135	97.75	512	86	
50%	149	55	130	1	153	98.7	515	91	
75%	223.5	61	140	1	166	99.65	519	94	
max	298	77	200	2	202	104.3	532	99	

Correlation Matrix

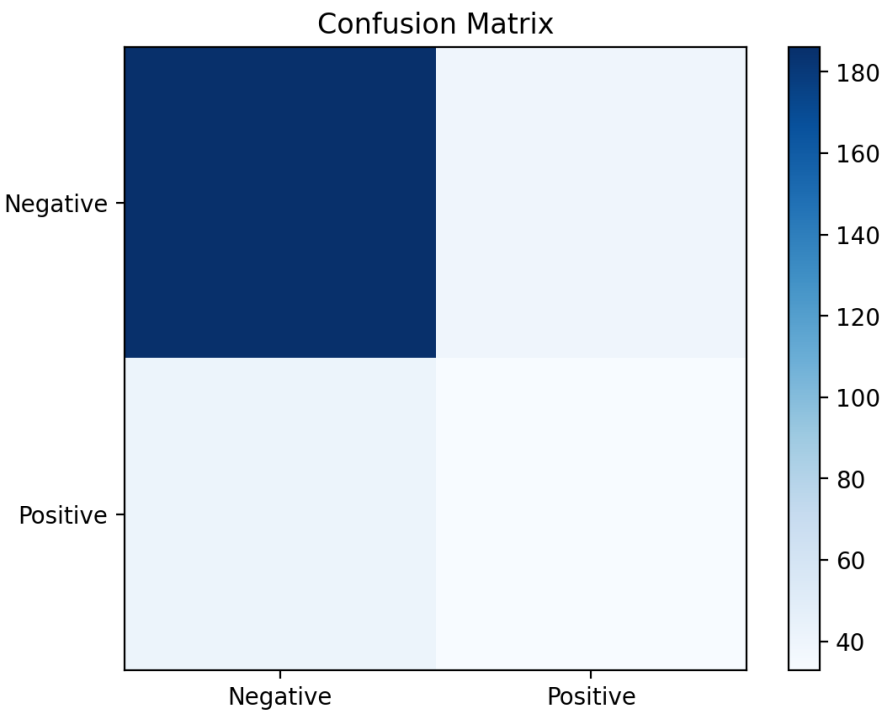


Modelling and results

The model that has been selected is: `Ensemble_ada-logistic-catboost.0.0.1`

This public information describes a model that consists of three different algorithms: logistic regression, Adaboost, and Catboost. These algorithms are combined together and the resulting classification is determined using a majority voting algorithm. This means that the majority of the three algorithms must agree on the classification in order for it to be

Confusion Matrix



accuracy: `0.7324414715719063`

precision: `0.4583333333333333`

recall: `0.44594594594594594`

f1: `0.4520547945205479`

Data descriptions

	accuracy	precision	recall	f1
0	0.7324	0.4583	0.4459	0.4521

Assumptions and Reasons

This assumption suggests that the heart rate values of unseen data should be expected to be similar to those observed in the training dataset. This means that the heart rate values of unseen data should be within the same range and have a similar distribution to the heart rate values in the training dataset. This assumption is useful when trying

Conclusion

Assumptions are made in data analysis to provide a more accurate representation of the data. In this case there is an assumption that data quality assurance can be improved and better models can be introduced to provide a more accurate representation of the data. This assumption is based on the fact that data quality assurance and better

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